KAIST

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[MAS456] Statistical Methods with Computer

Deciphering Patterns of Employment Discrimination and Health Outcomes

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Introduction

1.1 Research Background

Studying hiring discrimination in the labor market is crucial for social equity, economic efficiency, and organizational performance. This research focuses on the unique context of South Korea, using data from the Korean Labor and Income Panel Study (KLIPS). Globally, hiring discrimination, based on factors like gender, age, race, and ethnicity, hinders career opportunities and perpetuates socioeconomic disparities. South Korea's competitive job market, cultural specificities, and the prevalence of family-owned conglomerates add distinctive dimensions to the study. The KLIPS data, particularly from the 7th wave in 2004, offers valuable insights into hiring discrimination experiences. Employing advanced statistical techniques, this study (slightly) contributes to the literature by filling gaps in global discrimination studies and providing a methodological framework for a more comprehensive understanding of hiring discrimination factors. The findings have broader implications for informing policymakers, fostering inclusive workplace policies, and contributing to the global discourse on equitable hiring practices, addressing both empirical gaps and practical needs for a more equitable labor market.

1.2 Research Objective

- **Examine Demographic, Socio-Economic Factors and Hiring Discrimination's Association:**
  - **Objective:** Identify factors (age, gender, education level, income quartile, employment status) associated with hiring discrimination.
  - **Hypothesis:** Certain factors strongly correlate with hiring discrimination experiences in the South Korean labor market.
- **Identify Key Patterns Using Principal Component Analysis (PCA):**
  - **Objective:** Use PCA to reduce dataset complexity and identify principal components capturing significant variance in hiring discrimination responses.
  - **Hypothesis:** Principal components reveal distinct patterns related to hiring discrimination experiences and associated factors.
- **Segment Participants into Distinct Groups via Cluster Analysis:**
  - **Objective:** Categorize participants into clusters based on identified principal components to understand unique profiles within the workforce.
  - **Hypothesis:** Distinct subgroups exist within the workforce with unique hiring discrimination experiences and perceptions.
- **Develop Predictive Models for Hiring Discrimination Experiences:**
  - **Objective:** Construct and compare Random Forest and Penalized Logistic Regression models to predict hiring discrimination experiences, focusing on 'Not Applicable' responses.
  - **Hypothesis:** Predictive modeling can effectively estimate the likelihood of hiring discrimination, addressing potential under-reporting.
- **Investigate Correlation Between Hiring Discrimination and Self-Rated Health:**
  - **Objective:** Assess the association between hiring discrimination and self-rated health.
  - **Hypothesis:** Individuals experiencing hiring discrimination report differing levels of self-rated health compared to those who do not.

Method

2.1 Data Description

The dataset for this study is derived from the 7th wave (2004) of the Korean Labor and Income Panel Study (KLIPS), encompassing a sample of 3,576 waged workers. 3,479 people responded as either 'Yes' or 'No' but 97 people responded as 'Not Applicable'. The dataset is comprehensive, including a range of variables such as responses to hiring discrimination, gender, age, education level, marital status, employment status, income quartile, birth region, self-rated health, disability status, and residence location. Additionally, it encompasses various domains of discrimination experiences, providing a rich basis for analysis.

2.2 Principal Component Analysis (PCA)

PCA was employed to reduce the dimensionality of the data and to uncover underlying patterns. The approach involved:

- **Selection of Important PCs:** The PCA was conducted on the scaled data to ensure each variable contributed equally to the analysis. The number of components retained was determined based on the proportion of variance explained, with a focus on components that cumulatively explained a significant portion of the variance in the dataset. The first 6 PCs were found to explain more than 70% of the variance which is a reasonable threshold considering the graph of the variance coverage concerning the amount of PCs. The first 3 PCs accounted for around half of the variance, followed by subsequent PCs which added progressively smaller amounts.
- **Reporting and Interpretation:** Each identified principal component was analyzed to understand the weight and direction of each variable's contribution. This interpretation provided insights into the underlying constructs represented by the components, such as demographic factors, experiences of discrimination, or socio-economic status.

2.3 Principal Component Analysis (PCA) Findings

The PCA revealed several important principal components (PCs) that cumulatively explained a substantial portion of the variance in the dataset. We consider (Table 2) any loading value > 0.4 and any value < -0.4 to be Strong indicators. Key findings from the PCA include:

PC1:

Positive Loadings:	disc_jobedu, disc_promotion, disc_resign, disc_edu, disc_wage, emp_fin
Negative Loadings:	edu_cat, income_quartile
Interpretation:	PC1 captures experiences of discrimination and employment status. High values indicate discrimination and non-permanent employment, while low values suggest higher education and income.

PC2:

Positive Loadings:	edu_cat, disc_resign
Negative Loadings:	age, health
Interpretation:	PC2 contrasts education and discrimination in firing with age and health. Higher values suggest higher education and more firing-related discrimination in younger, healthier individuals.

PC3:

Positive Loadings:	disability
Negative Loadings:	gender
Interpretation:	PC3 is associated with disability and gender. High values may indicate a disability, while low values are associated with being female.

PC4:

Positive Loadings:	disability
Negative Loadings:	age, income_quartile
Interpretation:	PC4 is dominated by disability and contrasts with age and income. Higher values suggest a disability and lower income, while lower values indicate older age and higher income.

PC5:

Positive Loadings:	gender, health, disability
Negative Loadings:	None
Interpretation:	PC5 captures variance related to gender, self-rated health, and disability. High values suggest being female, worse health, and having a disability.

PC6:

Positive Loadings:	disc_edu
Negative Loadings:	disc_wage
Interpretation:	PC6 represents an axis contrasting discrimination in higher education with income level. Higher values suggest discrimination in education and lower income, while lower values suggest higher income and potentially less discrimination in education.

2.4 Cluster Analysis

Following PCA, cluster analysis was performed to segment the participants into distinct groups:

- **Selection of Number of Clusters:** The optimal number of clusters was determined using methods such as the elbow method and silhouette scores, ensuring that the chosen clusters effectively captured distinct patterns in the data. The elbow method did not show promising results so I decided to use silhouette scores to identify the correct number of clusters to use.

- **Characterization of Clusters:** Each cluster was characterized based on the mean values of the original variables and the scores on the principal components. This step provided a detailed description of the characteristics of each cluster, highlighting unique profiles within the workforce.

Cluster Analysis Findings

The cluster analysis of the dataset has yielded four distinct clusters considering both original data variables and principal components. Although 5 clusters looked slightly more promising for the Original Data, as the difference was minimal and for the sake of generalization, I decided to go with 4 clusters for both cases. Let's interpret each cluster in detail:

Clusters Based on Original Data:

Cluster 0 (Young, Highly Educated, Lower Discrimination Experiences):

Variable	Description
Gender	Slightly more males than females.
Age	Younger age group.
Education (edu_cat)	High educational attainment.
Employment (emp_fin)	Mostly permanent employees.
Income Quartile	High.
Health	Good self-rated health.
Experiences of Discrimination	Low in all categories.

This cluster represents a group of mostly young, highly educated individuals in stable employment, experiencing less discrimination and enjoying good health.

Cluster 1 (Older, Lower Education, Moderate Discrimination Experiences):

Variable	Description
Gender	Balanced gender distribution.
Age	Older age group.
Education	Lower educational levels.
Employment	Mix of permanent and non-permanent employees.
Income Quartile	Lower.
Health	Lower self-rated health.
Experiences of Discrimination	Moderate levels.

This cluster includes older individuals with lower education and moderate income, experiencing moderate levels of discrimination and reporting poorer health.

Cluster 2 (Middle-Aged, Educated, High Discrimination Experiences):

Variable	Description
Gender	Slightly more females than males.
Age	Middle-aged.
Education	Above-average education.
Employment	Mix of employment types.
Income Quartile	Average.
Health	Average self-rated health.
Experiences of Discrimination	High in all categories.

Members of this cluster are middle-aged, reasonably educated individuals facing significant discrimination in various areas.

Cluster 3 (Disability, High Discrimination in Some Areas):

Variable	Description
Gender	Predominantly male.
Age	Middle-aged.
Education	Moderate educational levels.
Employment	Mostly non-permanent employees.
Income Quartile	Lower.
Health	Poor self-rated health.
Disability	All have some form of disability.
Experiences of Discrimination	Particularly high in home and social activities.

This group consists of middle-aged males with disabilities, facing employment instability, lower income, and significant discrimination, particularly in non-work domains.

Clusters Based on Principal Components:

Cluster 0 (Low Discrimination, Higher Socio-Economic Status): Represents individuals likely experiencing lower levels of discrimination and enjoying higher socio-economic status, given the negative loadings on components related to discrimination and positive on socio-economic variables.
Cluster 1 (High Discrimination, Diverse Socio-Demographic Background): Indicates a group facing high levels of discrimination across various domains, with a diverse socio-demographic background.
Cluster 2 (Specific Discrimination Experiences, Possibly Health-Related): Likely represents individuals with specific experiences of discrimination, potentially related to health or disability issues, as indicated by the high loadings on related components.
Cluster 3 (Average Discrimination and Socio-Economic Status): Suggests a group with average levels of discrimination experiences and socio-economic status.

2.5 Predictive Modeling

Two predictive models were considered for forecasting hiring discrimination experiences:

- **Random Forest and Penalized Logistic Regression with Lasso:**
  - **Random Forest:** A non-linear model known for its robustness and ability to handle complex interactions among variables. The model was tuned for parameters like the number of trees and depth of trees.
  - **Penalized Logistic Regression with Lasso:** A linear approach with a Lasso penalty to encourage sparsity, thereby selecting a subset of variables. The tuning focused on the regularization strength.
- **Model Comparison:** Both models were compared based on their AUC scores from the ROC analysis, providing a clear metric for predictive performance.
- **Selection of Best Model and Threshold:** The model with the higher AUC was selected as the best prediction model. Subsequently, the optimal threshold for binary prediction was determined from the ROC curve, aimed at maximizing the sum of sensitivity and specificity.

Predictive Modeling Findings

The comparison between the Random Forest and Penalized Logistic Regression models revealed distinct insights:

- **Model Performance:**
  - The AUC of the ROC for the Random Forest model was reported, indicating its ability to distinguish between the classes. While we saw an 89.63% AUC for logistic regression, the AUC for the Random Forest model was 91.5%.
  - The Penalized Logistic Regression model's AUC was slightly lower, suggesting a somewhat lesser but still significant predictive capability.
- **Best Model Selection:** Based on the AUC scores, the Random Forest model was selected as the more effective model for predicting hiring discrimination experiences. I have also tried tuning the model with various parameters. The best results were obtained for max_depth=5, and n_estimators=300.
- **Optimal Threshold for Binary Prediction:** The threshold derived from the ROC curve for the Random Forest model was identified, ensuring the best balance between sensitivity and specificity. The value turned out to be 0.18278975364991587.

Result

The results indicate that hiring discrimination in the South Korean labor market is a multi-faceted issue, closely intertwined with various socio-demographic factors. The predictive models further elucidate potential under-reporting in responses related to hiring discrimination, with significant implications for understanding the true extent of discriminatory practices in the labor market.

3.1 Question 1

We have fit a logistical regression model in the dataset to find out the variables that are significant in determining hiring discrimination. We assume any variable with < 0.05 p-value to be significant.

Variable	coef	P$>|z|$	Variable	coef	P$>|z|$	Variable	coef	P$>|z|$
gender	-0.0521	0.664	disc_jobedu	0.1118	0.325	disc_promotion	0.1990	0.063
age	0.1300	0.037	birth_region	-0.1042	0.484	disc_resign	0.1593	0.100
edu_cat	-0.0691	0.454	health	0.1810	0.037	disc_edu	-0.0154	0.914
mariage	-0.4246	0.001	disability	0.2605	0.429	disc_home	-0.1022	0.677
emp_fin	0.4897	0.000	residence	0.0396	0.001	disc_social	1.1763	0.000
income_quartile	-0.2322	0.000	disc_wage	3.2172	0.000	-	-	-

Table 1: Logit Regression Results

The variables that turned out to be important that are associated with the experience of hiring discrimination are age, marriage, emp_fin, income_quartile, health, residence, disc_wage, and disc_social.

Age

Age (coef: 0.130, p-value: 0.037): Age has a positive coefficient, suggesting that older individuals are more likely to report experiencing hiring discrimination. The relationship is statistically significant.

Marital Status

Marital Status (marriage, coef: -0.4246, p-value: 0.001): Being married is negatively associated with the experience of hiring discrimination, meaning married individuals are less likely to report such experiences. This relationship is statistically significant.

Employment Status

Employment Status (emp_fin, coef: 0.4897, p-value: 0.000): Non-permanent employees are more likely to experience hiring discrimination compared to permanent employees. This is a significant association.

Income Quartile

Income Quartile (income_quartile, coef: -0.2322, p-value: 0.000): Higher income quartiles are negatively associated with hiring discrimination experiences, suggesting that individuals in higher income brackets are less likely to report discrimination.

Self-rated Health

Self-rated Health (health, coef: 0.1810, p-value: 0.037): A better self-rated health status is associated with a higher likelihood of experiencing hiring discrimination, which is statistically significant.

Residence

Residence (residence, coef: 0.0396, p-value: 0.001): This suggests a significant relationship between the area of residence and the experience of hiring discrimination, with certain regions potentially having higher incidences.

Experience of Discrimination in Receiving Income

Experience of Discrimination in Receiving Income (disc_wage, coef: 3.2172, p-value: 0.000): This has a strong positive association with hiring discrimination, indicating that those who have experienced wage discrimination are much more likely to report hiring discrimination.

Experience of Discrimination in General Social Activities

Experience of Discrimination in General Social Activities (disc_social, coef: 1.1763, p-value: 0.000): Similar to disc_wage, experiencing discrimination in social activities is significantly and positively associated with hiring discrimination.

Other variables like gender, education level, birth region, disability, and experiences of discrimination in other aspects (job education, promotion, resignation, education, and home) were not found to be significantly associated with hiring discrimination in this model.

3.2 Question 2

The principal components (PCs) derived from the analysis of 12 explanatory variables (gender, age, education level, employment status, income, self-rated health, disability, and experiences of discrimination in various contexts) can be interpreted as follows:

- **PC1:** Captures experiences of discrimination across various domains and employment status. High values indicate strong reported discrimination and non-permanent employment, while low values suggest higher education and income.
- **PC2:** Contrasts educational attainment and discrimination in firing with age and health. Higher values suggest higher education, more reported discrimination in firing, and younger, healthier individuals. Lower values indicate older age and poorer health.
- **PC3:** Associated with disability and gender. High values may indicate the presence of a disability, while low values are associated with being female.
- **PC4:** Dominated by disability status, contrasting with age and income. Higher values suggest the presence of a disability and lower income, while lower values indicate older age and higher income.
- **PC5:** Captures variance primarily related to gender, self-rated health, and disability. High values suggest a higher likelihood of being female, having worse self-rated health, and having a disability.
- **PC6:** Represents a contrast between discrimination in higher education and income level. Higher values suggest experiences of discrimination in education and lower income, whereas lower values suggest higher income and potentially less discrimination in education.

In summary, the principal components capture different aspects of discrimination, employment status, socio-economic status, and demographic characteristics. The PCs provide a condensed representation of the data where each component represents a combination of features that together account for different portions of the data's variance.

3.3 Question 3

Following Principal Component Analysis (PCA) on 12 variables, cluster analysis was performed to segment participants into distinct groups. The clustering was done based on both the original data variables and the scores on the principal components.

Clusters Based on Original Data Variables:

1. **Cluster 0: Young, Highly Educated, Lower Discrimination Experiences** - Comprised mostly of young, well-educated individuals in stable employment, experiencing less discrimination and enjoying good health.
2. **Cluster 1: Older, Lower Education, Moderate Discrimination Experiences** - Includes older individuals with lower education and moderate-income, experiencing moderate levels of discrimination and poorer health.
3. **Cluster 2: Middle-Aged, Educated, High Discrimination Experiences** - Consists of middle-aged, reasonably educated individuals facing significant discrimination in various areas.
4. **Cluster 3: Disability, High Discrimination in Some Areas** - Group of middle-aged males with disabilities, facing employment instability, lower income, and significant discrimination, particularly in non-work domains.

Clusters Based on Principal Components:

1. **Cluster 0: Low Discrimination, Higher Socio-Economic Status** - Represents individuals likely experiencing lower levels of discrimination and enjoying higher socio-economic status.
2. **Cluster 1: High Discrimination, Diverse Socio-Demographic Background** - Indicates a group facing high levels of discrimination across various domains, with a diverse socio-demographic background.
3. **Cluster 2: Specific Discrimination Experiences, Possibly Health-Related** - Likely represents individuals with specific experiences of discrimination, potentially related to health or disability issues.
4. **Cluster 3: Average Discrimination and Socio-Economic Status** - Suggests a group with average levels of discrimination experiences and socio-economic status.

Comparison of Clustering Results:

The clusters based on original data variables highlight differences primarily in terms of age, education, health, and discrimination experiences. In contrast, the clusters derived from the principal components focus more on the levels of discrimination and socio-economic status. The PCA-based clusters offer a more nuanced understanding of the interplay between socio-economic factors and experiences of discrimination. This comparison reveals the multifaceted nature of the workforce, indicating diverse experiences related to discrimination, socio-economic status, health, and demographics.

3.4 Question 4

predicted_disc_hire	0	1
gender		
Male (0)	0.515625	0.484375
Female (1)	0.090909	0.909091

The predicted distribution of responses for the experience of hiring discrimination (`predicted_disc_hire`) between males (`gender = 0`) and females (`gender = 1`) suggests a difference in reporting between the two genders among the 97 participants who originally responded with 'Not Applicable':

- For males, the predicted probabilities are split almost evenly, with 51.56% predicted to respond 'No' and 48.44% predicted to respond 'Yes' to having experienced hiring discrimination.
- For females, a significant majority, 90.91%, are predicted to respond 'Yes' to having experienced hiring discrimination, with only 9.09% predicted to respond 'No'.

This disparity indicates that, according to the model's predictions, females in this subset are much more likely to report having experienced hiring discrimination compared to males. This could point towards a potential under-reporting of hiring discrimination among males, or it could indicate a higher prevalence of discrimination experiences among females.

3.5 Question 5

The ANOVA test results show that there is a statistically significant difference in self-rated health among the four groups, as indicated by the very small p-value (2.04×10^{-6}).
The subsequent Tukey HSD post-hoc test results provide pairwise comparisons between the groups. Here's the interpretation of the results:

- **NA_No vs. NA_Yes:** No significant difference in self-rated health ($p = 0.9526$).
- **NA_No vs. No:** No significant difference in self-rated health ($p = 0.9744$).
- **NA_No vs. Yes:** No significant difference in self-rated health ($p = 0.8013$).
- **NA_Yes vs. No:** No significant difference in self-rated health ($p = 0.4964$).
- **NA_Yes vs. Yes:** No significant difference in self-rated health ($p = 0.9874$).
- **No vs. Yes:** A significant difference in self-rated health ($p = 0.0$), with the mean difference being 0.1534.

The significant result in the last comparison indicates that individuals who answered 'Yes' to experiencing hiring discrimination have a different self-rated health status than those who answered 'No'. Specifically, there seems to be a higher self-rated health score among those who did not report experiencing hiring discrimination.

For the individuals who initially answered 'NA' but were predicted to have answered 'No' or 'Yes', their self-rated health does not significantly differ from those who explicitly answered 'No' or 'Yes'. This suggests that the prediction model used to estimate the responses of the 'NA' group is consistent with the overall pattern of self-rated health associated with hiring discrimination experiences.

In conclusion, there is an association between the experience of hiring discrimination and self-rated health among the original groups ('No' vs. 'Yes'), but not among the 'NA' group when classified into 'No' or 'Yes' based on the prediction model.

Conclusions

This research on hiring discrimination in the South Korean labor market, utilizing data from the Korean Labor and Income Panel Study (KLIPS), reveals significant findings and their implications.

4.1 Summary of Key Findings

- **Demographic and Socio-Economic Associations**: Logistic regression highlights significant links between hiring discrimination and factors such as age, marital status, employment status, income quartile, and self-rated health.
- **Principal Component Analysis (PCA)**: PCA identifies principal components reflecting discrimination experiences and socio-economic factors, emphasizing the multifaceted nature of hiring discrimination.
- **Cluster Analysis**: Reveals diverse workforce clusters, ranging from those experiencing lower discrimination to those facing significant discrimination.
- **Predictive Modeling**: The Random Forest model effectively predicts hiring discrimination experiences, indicating a notable gender disparity in reporting.

4.2 Implications and Recommendations

- **Policy Interventions**: Findings advocate for policy reforms focusing on mitigating hiring discrimination, especially concerning age, marital status, and income.
- **Workplace Inclusion**: Diverse workforce profiles call for tailored inclusion strategies.
- **Awareness Programs**: Gender disparities in reporting discrimination underline the need for awareness and educational initiatives.

4.3 Limitations and Future Directions

- **Cross-Sectional Data**: Future studies should consider longitudinal approaches for causal inferences.
- **Cultural Generalizability**: Comparative international studies are recommended to broaden the findings' applicability.
- **Qualitative Insights**: Incorporating qualitative research can offer deeper understanding of individual experiences.

In summary, this study sheds light on the dynamics of hiring discrimination in South Korea, underscoring the need for targeted policies and inclusive workplace practices. Future research should expand upon these findings, exploring longitudinal trends and incorporating qualitative methods for a more holistic understanding.

Appendix

Variable	PC1	PC2	PC3	PC4	PC5	PC6
gender	0.167435	0.016164	-0.791322	0.043433	0.420158	-0.208187
age	0.322990	-0.596856	0.336981	-0.404556	-0.060399	-0.204470
edu_cat	-0.512316	0.596214	0.037254	0.090483	0.126472	0.204848
emp_fin	0.430839	-0.377448	-0.342500	0.225503	-0.220718	0.018825
income_quartile	-0.443912	0.328414	0.136127	-0.407821	0.348345	-0.389241
health	0.272044	-0.494769	-0.041915	-0.221880	0.521121	0.237606
disability	0.143707	-0.201233	0.473983	0.686229	0.440559	-0.143829
disc_wage	0.581898	0.194007	0.095080	-0.017882	-0.024730	-0.405027
disc_jobedu	0.769294	0.312607	0.007479	-0.020416	-0.065548	-0.044124
disc_promotion	0.767150	0.349258	-0.004350	0.007794	-0.058350	-0.039403
disc_resign	0.559861	0.466811	0.109063	-0.014171	0.088114	0.019949
disc_edu	0.536045	0.158167	0.140313	-0.221308	0.221716	0.503064

Table 2: Loadings for Principal Components

Table 3: Part 1 of the Centroids for Original Data

cluster	gender	age	edu_cat	emp_fin	income_quartile	health	disability
0	0.350000	1.482828	1.581818	0.057576	2.175253	1.201010	1.387779e-17
1	0.437963	2.797222	0.463889	0.464815	1.239815	1.687963	1.040834e-17
2	0.542453	2.193396	0.740566	0.466981	1.167453	1.523585	1.040834e-17
3	0.163043	2.641304	0.630435	0.369565	1.108696	1.750000	1.000000e+00

Table 4: Part 2 of the Centroids for Original Data

disc_wage	disc_jobedu	disc_promotion	disc_resign	disc_edu
0.079798	0.024747	0.101010	0.153030	0.018182
0.174074	0.041667	0.100926	0.045370	0.043519
0.639151	1.610849	1.754717	1.183962	0.580189
0.326087	0.347826	0.456522	0.402174	0.206522

Cluster	PC1	PC2	PC3	PC4	PC5	PC6
0	0.28	-1.40	-0.26	-0.22	-0.26	-0.07
1	3.55	1.28	-0.07	-0.25	-0.14	0.12
2	1.53	-1.64	3.12	4.14	2.57	-0.78
3	-1.01	0.54	0.01	-0.02	0.05	0.05

Table 5: Centroids for Principal Components

Group 1	Group 2	Mean Diff.	P-Adj.	Lower	Upper	Reject
NA_No	NA_Yes	0.0751	0.9526	-0.2915	0.4417	False
NA_No	No	-0.0482	0.9744	-0.3408	0.2443	False
NA_No	Yes	0.1052	0.8013	-0.193	0.4034	False
NA_Yes	No	-0.1234	0.4964	-0.3491	0.1024	False
NA_Yes	Yes	0.0301	0.9874	-0.203	0.2631	False
No	Yes	0.1534	0.0	0.0791	0.2278	True

Table 6: Multiple Comparison of Means - Tukey HSD, FWER=0.05

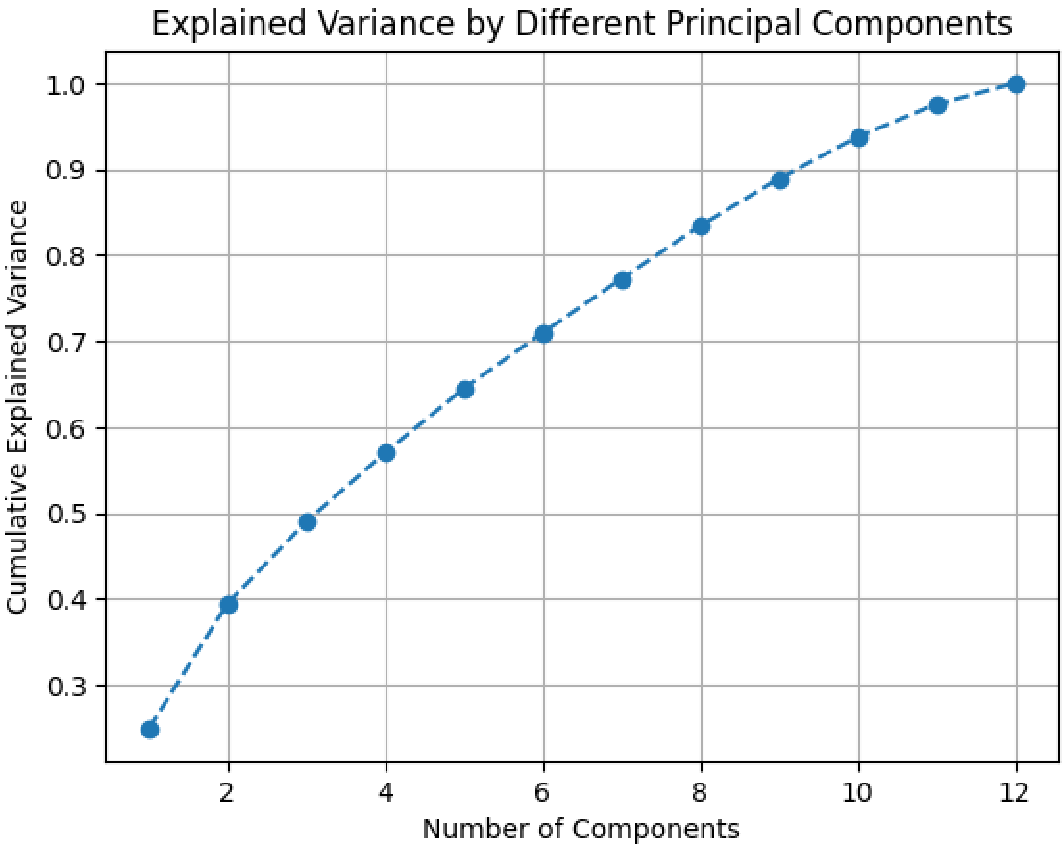

Figure 1: Explained Variance by Different Principal Components

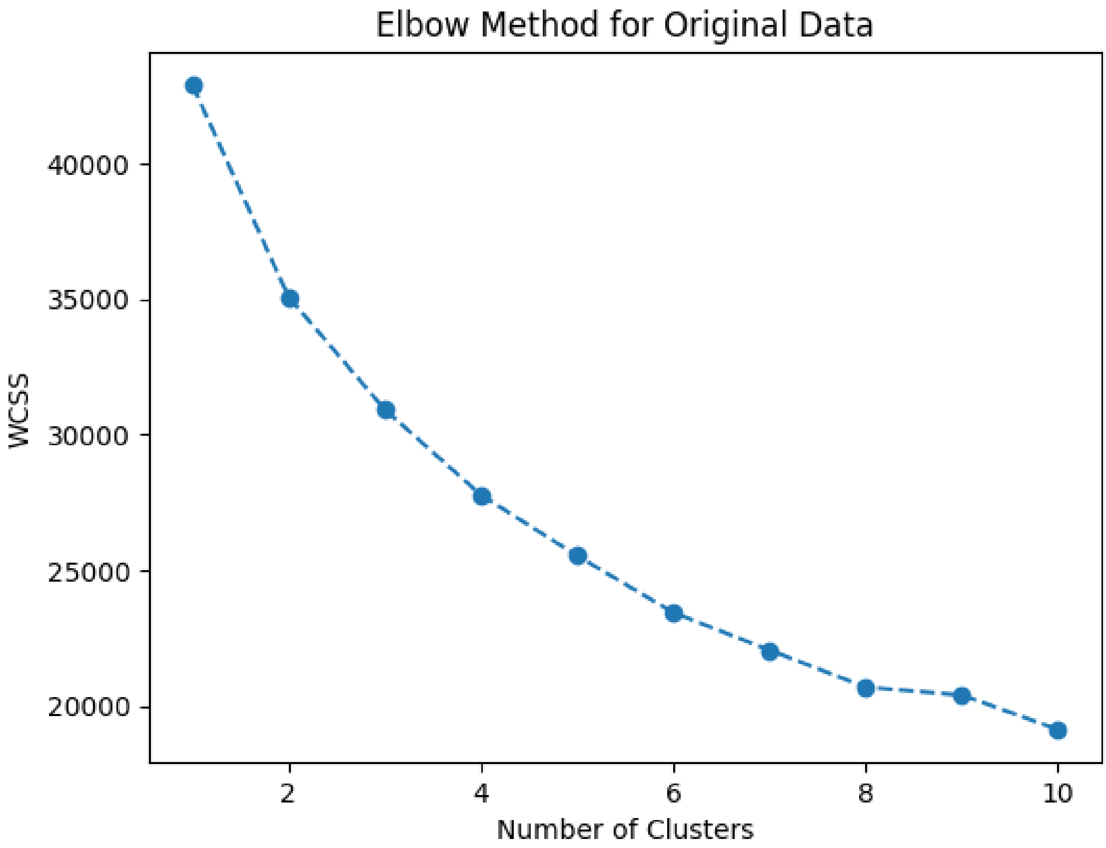

Figure 2: Elbow Method for Original Data

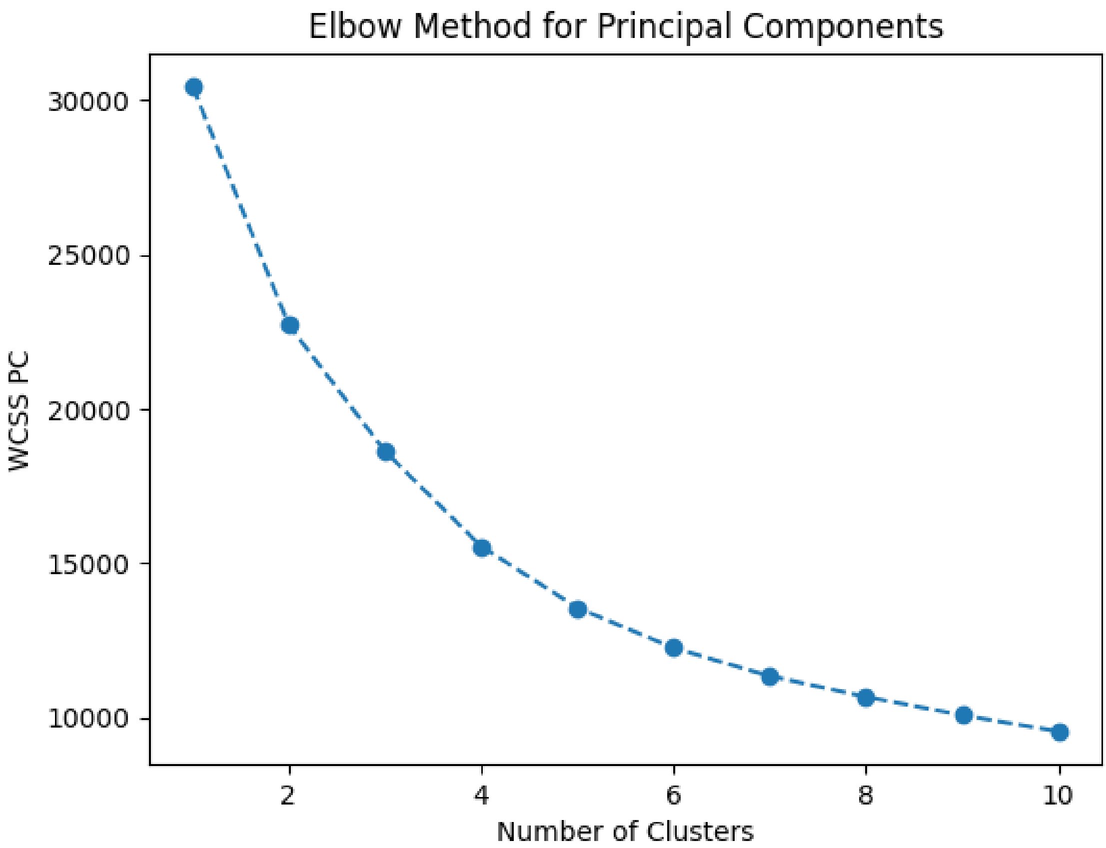

Figure 3: Elbow Method for Principal Components

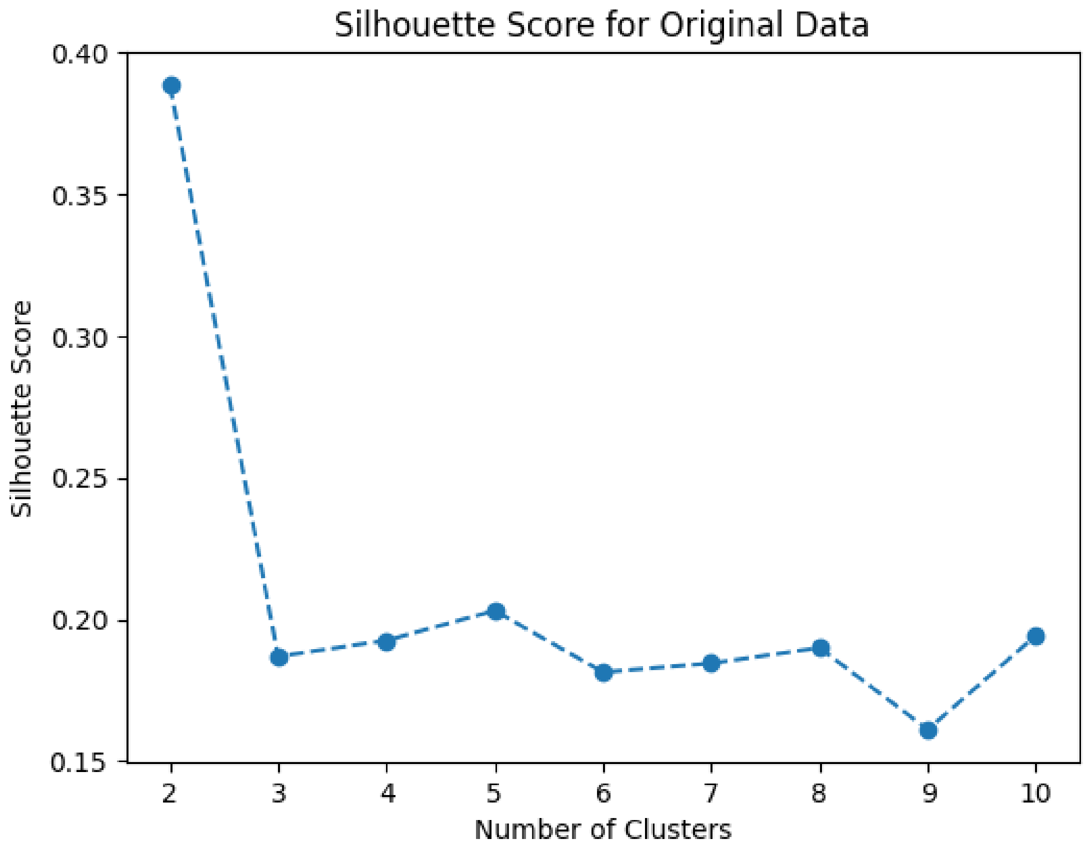

Figure 4: Sillouette Score for Original Data

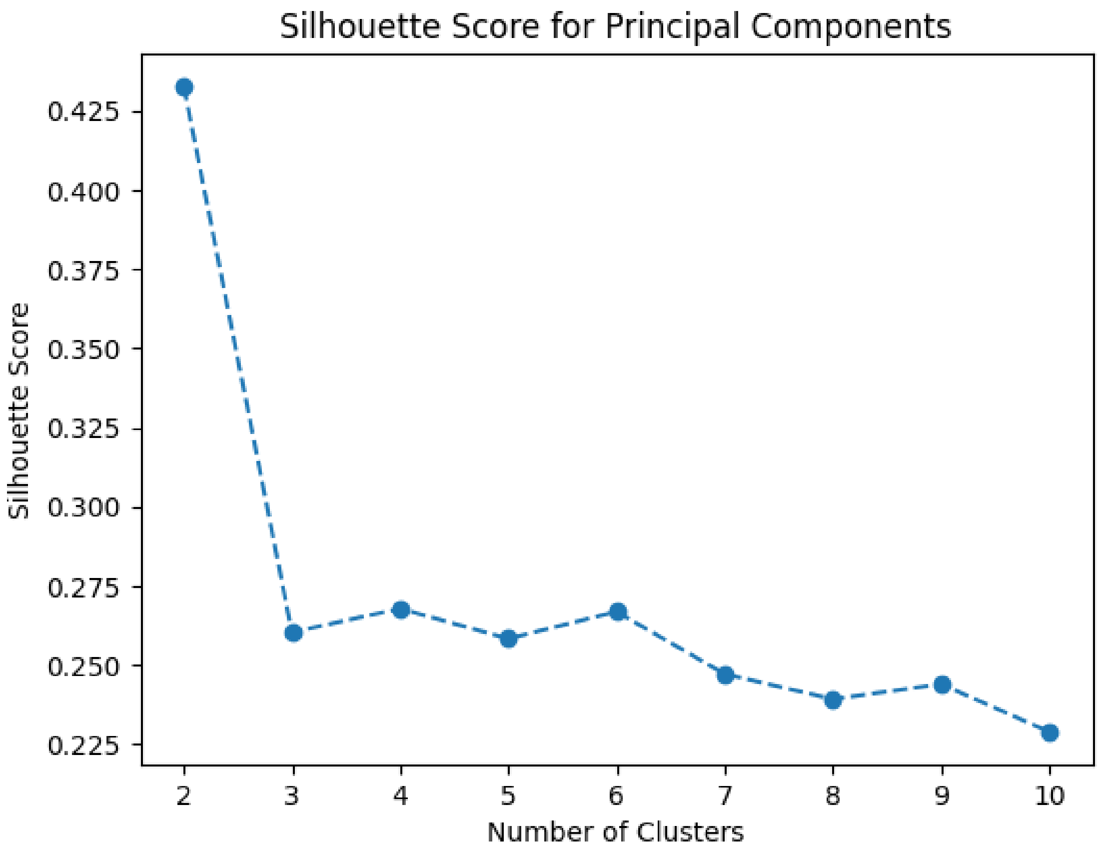

Figure 5: Sillouette Score for Principal Components

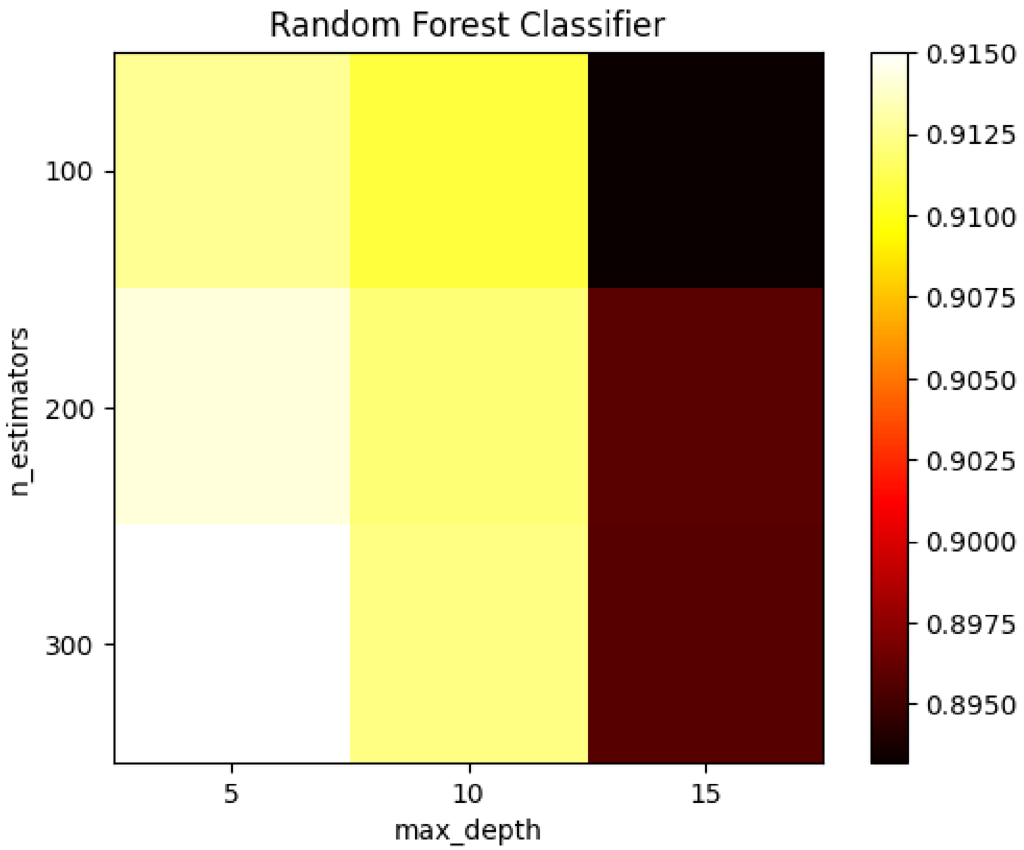

Figure 6: Random Forest Classifier

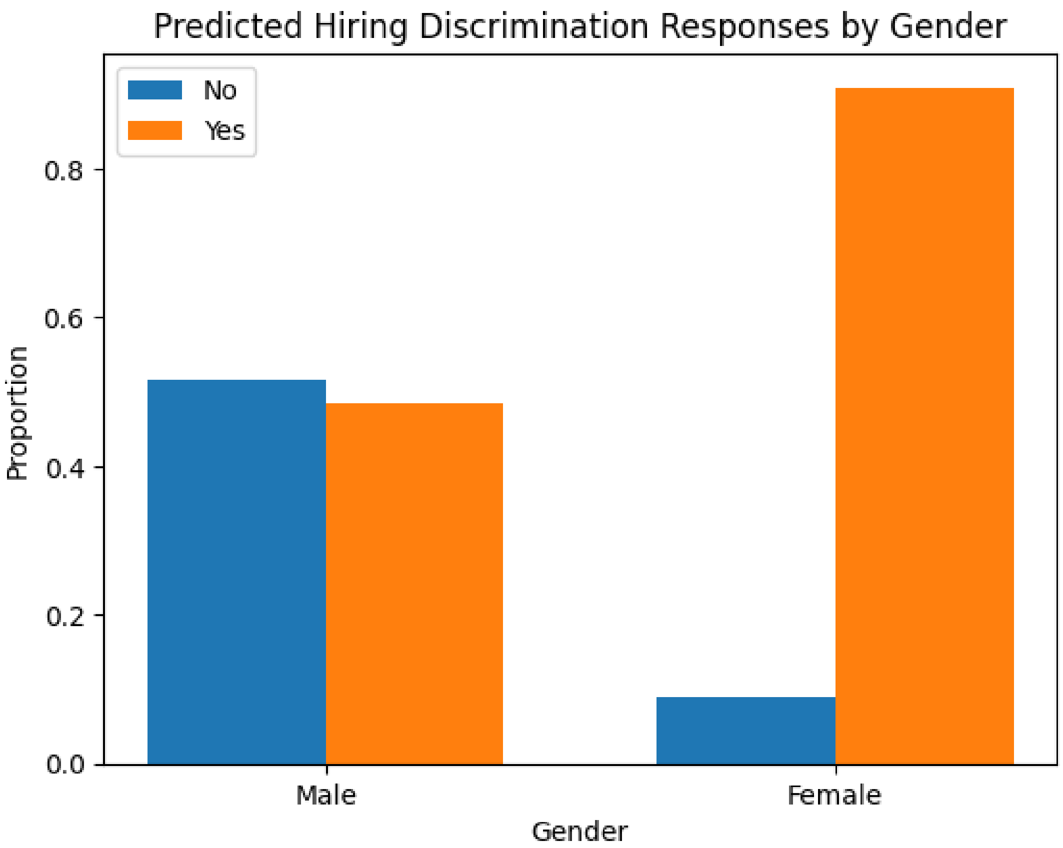

Figure 7: Difference in prediction based on Gender

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